

CIND 123

Data Analytics: Basic Methods - Assignment 3 (10%)

Zion Yitbarek

Section DJ0, Student Number: 500883564

Instructions

This assignment can be submitted in either Python or R, as preferred. For R submissions, ensure you submit both an RMD file and a knitted PDF/HTML file. Comment out any `install.packages` commands to prevent errors during knitting.

NOTE: If you choose R, all responses should be in R (ignore Python questions). Likewise, if you use Python, all answers should be in Python.

Question 1 (25 points)

Midterm and final grades of some students are provided. Set up the variables `midterm` and `final`.

```
# Data setup
midterm <- c(87, 72, 80, 91, 85, 69, 40, 34, 61)
final <- c(59, 68, 52, 62, 78, 70, 35, 28, 45)
```

Q1a (10 points)

Find the least-squares line relating `final` to `midterm`.

Does the assumption of a linear relationship appear to be reasonable in this case? Give reasons to your answer as a comment.

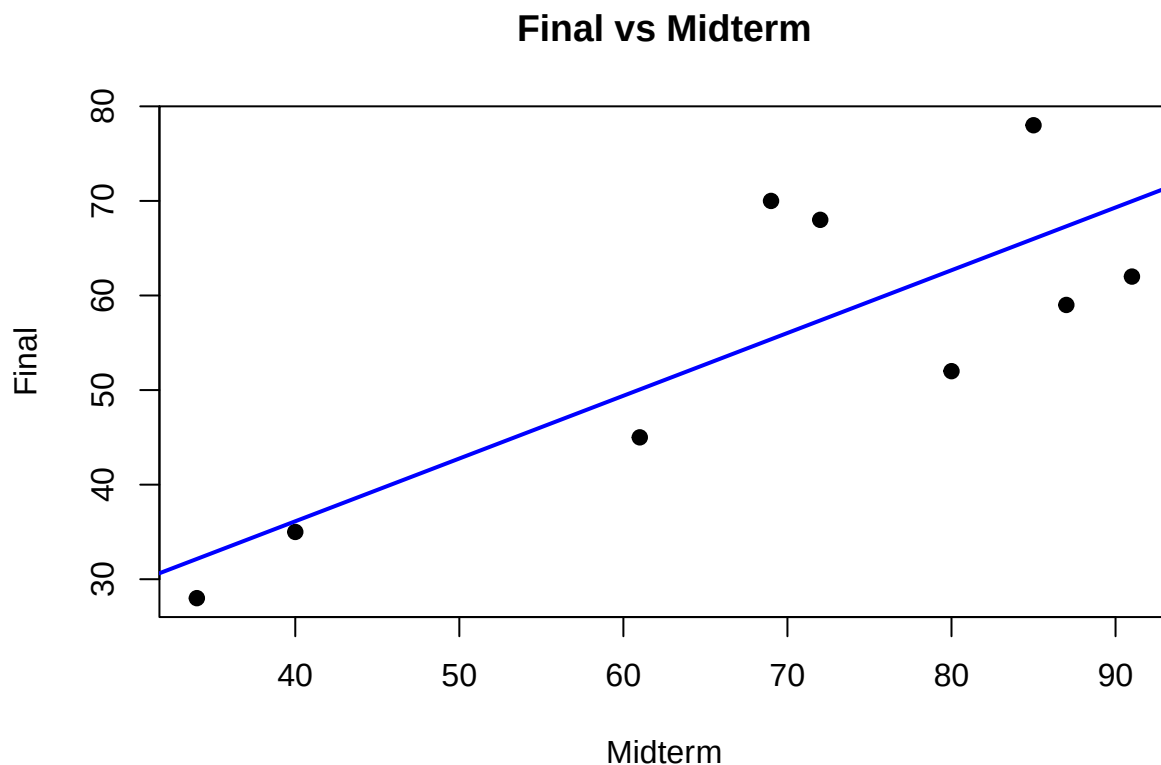
```
midterm <- c(87, 72, 80, 91, 85, 69, 40, 34, 61)
final <- c(59, 68, 52, 62, 78, 70, 35, 28, 45)
```

```
model <- lm(final ~ midterm)
summary(model)
```

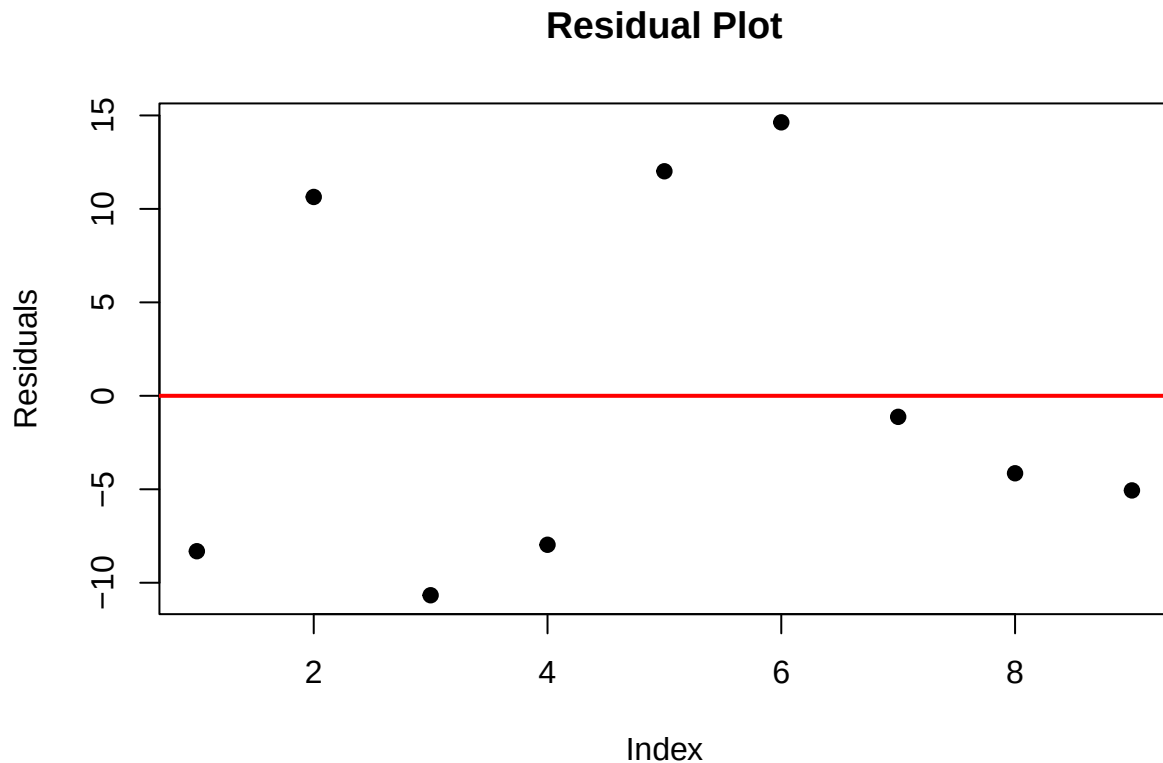
```
##
## Call:
## lm(formula = final ~ midterm)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.669  -7.968  -4.144   10.640   14.630
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.5827    12.9246   0.741   0.483
## midterm       0.6636     0.1810   3.666   0.008 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.43 on 7 degrees of freedom
## Multiple R-squared:  0.6576, Adjusted R-squared:  0.6087
## F-statistic: 13.44 on 1 and 7 DF, p-value: 0.008002
```

```
plot(midterm, final, main="Final vs Midterm", xlab="Midterm", ylab="Final", pch=19)
abline(model, col="blue", lwd=2) # Add regression line
```



```
plot(model$residuals, main="Residual Plot", ylab="Residuals", xlab="Index", pch=19)
abline(h=0, col="red", lwd=2) # Reference line at zero
```



Comment:

The scatter plot is showing an upward trend. This indicates that there is a positive relationship between the variables.

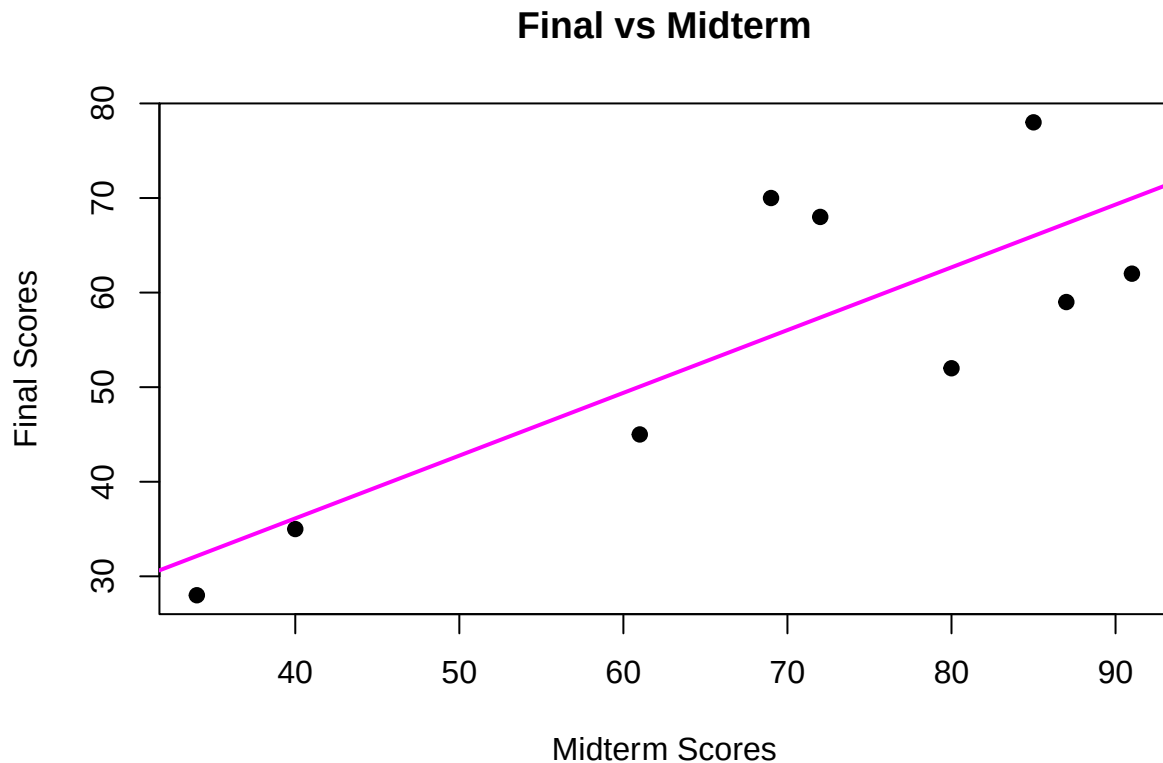
The residual plot shows random scatter with the line of best fit exactly at zero. This means that a linear model is appropriate for the data.

Q1b (10 points)

Plot the `final` as a function of the `midterm` using a scatterplot and graph the least-square line in red color on the same plot.

```
# Scatter plot of final vs midterm
plot(midterm, final,
     main="Final vs Midterm",
     xlab="Midterm Scores",
     ylab="Final Scores",
     pch=19)

# Add regression line
abline(model, col="magenta", lwd=2)
```



Q1c (5 points)

Use the regression line to predict final grades when midterm grades are 82 and 32.

```
# Predict final grades for midterm scores of 82 and 32
new_data1 <- data.frame(midterm = c(82, 32))
predicted_final <- predict(model, new_data1)

# Print predictions
print(predicted_final)
```

```
##           1           2
## 63.99622 30.81723
```

Question 2 (50 points)

This question uses the `plm` package. Load the Crime dataset as follows:

```
# Crime = read.csv("crime.csv")
# install.package(plm) # If needed, comment out this line before knitting
library(plm)
```

```
## Warning: package 'plm' was built under R version 4.4.3
```

```
data("Crime")
```

Q2a (5 points)

Display the first 7 rows of `crime` data and display the names of all the variables, the number of variables, then display a descriptive summary of each variable.

```
library(plm)
data("Crime")
head(Crime, 7)
```

```
##      county year      crmrte  prbarr  prbconv  prbpris avgsgen      polpc  density
## 1         1   81 0.0398849 0.289696 0.402062 0.472222   5.61 0.0017868 2.307159
## 2         1   82 0.0383449 0.338111 0.433005 0.506993   5.59 0.0017666 2.330254
## 3         1   83 0.0303048 0.330449 0.525703 0.479705   5.80 0.0018358 2.341801
## 4         1   84 0.0347259 0.362525 0.604706 0.520104   6.89 0.0018859 2.346420
## 5         1   85 0.0365730 0.325395 0.578723 0.497059   6.55 0.0019244 2.364896
## 6         1   86 0.0347524 0.326062 0.512324 0.439863   6.90 0.0018952 2.385681
## 7         1   87 0.0356036 0.298270 0.527596 0.436170   6.71 0.0018279 2.422633
##      taxpc  region smsa  pctmin      wcon      wtuc      wtrd      wfir      wser
## 1 25.69763 central   no 20.2187 206.4803 333.6209 182.3330 272.4492 215.7335
## 2 24.87425 central   no 20.2187 212.7542 369.2964 189.5414 300.8788 231.5767
## 3 26.45144 central   no 20.2187 219.7802 1394.8030 196.6395 309.9696 240.1568
## 4 26.84235 central   no 20.2187 223.4238 398.8604 200.5629 350.0863 252.4477
## 5 28.14034 central   no 20.2187 243.7562 358.7830 206.8827 383.0707 261.0861
## 6 29.74098 central   no 20.2187 257.9139 369.5465 218.5165 409.8842 269.6129
## 7 30.99368 central   no 20.2187 281.4259 408.7245 221.2701 453.1722 274.1775
##      wmfgr  wfed  wsta  wloc      mix  pctymle  lcrmrte  lprbarr
## 1 229.12 409.37 236.24 231.47 0.0999179 0.0876968 -3.221757 -1.238923
## 2 240.33 419.70 253.88 236.79 0.1030491 0.0863767 -3.261134 -1.084381
## 3 269.70 438.85 250.36 248.58 0.0806787 0.0850909 -3.496449 -1.107303
## 4 281.74 459.17 261.93 264.38 0.0785035 0.0838333 -3.360270 -1.014662
## 5 298.88 490.43 281.44 288.58 0.0932486 0.0823065 -3.308445 -1.122715
## 6 322.65 478.67 286.91 306.70 0.0973228 0.0800806 -3.359507 -1.120668
## 7 334.54 477.58 292.09 311.91 0.0801688 0.0778710 -3.335309 -1.209756
##      lprbconv  lprbpris  lavgsen  lpolpc  ldensity  lwcon  lwtuc  lwtrd
## 1 -0.9111490 -0.7503061 1.724551 -6.327340 0.8360171 5.330205 5.810005 5.205835
## 2 -0.8370060 -0.6792581 1.720979 -6.338704 0.8459773 5.360137 5.911600 5.244607
## 3 -0.6430188 -0.7345839 1.757858 -6.300291 0.8509204 5.392628 7.240509 5.281372
## 4 -0.5030129 -0.6537265 1.930071 -6.273361 0.8528909 5.409070 5.988612 5.301128
## 5 -0.5469313 -0.6990466 1.879465 -6.253162 0.8607340 5.496169 5.882718 5.332152
## 6 -0.6687981 -0.8212920 1.931521 -6.268420 0.8694848 5.552626 5.912277 5.386862
## 7 -0.6394244 -0.8297232 1.903599 -6.304609 0.8848549 5.639869 6.013041 5.399384
##      lwfir  lwser  lwmfgr  lwfed  lwsta  lwloc  lpctymle  lpctmin
## 1 5.607452 5.374044 5.434246 6.014619 5.464848 5.444450 -2.433870 3.006608
## 2 5.706707 5.444911 5.482013 6.039540 5.536862 5.467174 -2.449038 3.006608
## 3 5.736475 5.481292 5.597310 6.084157 5.522900 5.515765 -2.464036 3.006608
## 4 5.858180 5.531204 5.640985 6.129421 5.568077 5.577387 -2.478925 3.006608
## 5 5.948220 5.564850 5.700042 6.195282 5.639919 5.664972 -2.497306 3.006608
## 6 6.015875 5.596987 5.776568 6.171011 5.659169 5.725870 -2.524721 3.006608
## 7 6.116272 5.613776 5.812757 6.168732 5.677062 5.742715 -2.552702 3.006608
##      ltaxpc      lmix
```

```
## 1 3.246399 -2.303407
## 2 3.213833 -2.272549
## 3 3.275311 -2.517281
## 4 3.289981 -2.544612
## 5 3.337204 -2.372487
## 6 3.392526 -2.329722
## 7 3.433783 -2.523621
```

```
names(Crime)
```

```
## [1] "county" "year" "crmte" "prbarr" "prbconv" "prbpris"
## [7] "avgsen" "polpc" "density" "taxpc" "region" "smsa"
## [13] "pctmin" "wcon" "wtuc" "wtrd" "wfir" "wser"
## [19] "wmfg" "wfed" "wsta" "wloc" "mix" "pctymle"
## [25] "lcrmte" "lprbarr" "lprbconv" "lprbpris" "lavgsen" "lpolpc"
## [31] "ldensity" "lwcon" "lwtuc" "lwtrd" "lwfir" "lwser"
## [37] "lwmfg" "lwfed" "lwsta" "lwloc" "lpctymle" "lpctmin"
## [43] "ltaxpc" "lmix"
```

```
ncol(Crime)
```

```
## [1] 44
```

```
summary(Crime)
```

```
##      county      year      crmte      prbarr
## Min.   : 1.0   Min.   :81   Min.   :0.001812   Min.   :0.05882
## 1st Qu.: 51.0   1st Qu.:82   1st Qu.:0.018352   1st Qu.:0.21790
## Median :103.0   Median :84   Median :0.028441   Median :0.27824
## Mean   :100.6   Mean    :84   Mean    :0.031588   Mean    :0.30737
## 3rd Qu.:151.0   3rd Qu.:86   3rd Qu.:0.038406   3rd Qu.:0.35252
## Max.   :197.0   Max.    :87   Max.    :0.163835   Max.    :2.75000
##      prbconv      prbpris      avgsen      polpc
## Min.   : 0.06838   Min.   :0.1489   Min.   : 4.220   Min.   :0.0004585
## 1st Qu.: 0.34769   1st Qu.:0.3744   1st Qu.: 7.160   1st Qu.:0.0011913
## Median : 0.47437   Median :0.4286   Median : 8.495   Median :0.0014506
## Mean   : 0.68862   Mean    :0.4255   Mean    : 8.955   Mean    :0.0019168
## 3rd Qu.: 0.63560   3rd Qu.:0.4832   3rd Qu.:10.197   3rd Qu.:0.0018033
## Max.   :37.00000   Max.    :0.6786   Max.    :25.830   Max.    :0.0355781
##      density      taxpc      region      smsa      pctmin
## Min.   : 0.1977   Min.   : 14.30   other :245   no :574   Min.   : 1.284
## 1st Qu.:0.5329   1st Qu.: 23.43   west  :147   yes: 56   1st Qu.:10.005
## Median :0.9526   Median : 27.79   central:238   Median :24.852
## Mean   :1.3861   Mean    : 30.24   Mean    :25.713
## 3rd Qu.:1.5078   3rd Qu.: 33.27   3rd Qu.:38.223
## Max.   :8.8277   Max.    :119.76   Max.    :64.348
##      wcon      wtuc      wtrd      wfir
## Min.   : 65.62   Min.   : 28.86   Min.   : 16.87   Min.   : 3.516
## 1st Qu.:201.66   1st Qu.:317.60   1st Qu.:168.05   1st Qu.:235.705
## Median :236.46   Median :358.20   Median :185.48   Median :264.423
## Mean   :245.67   Mean    :406.10   Mean    :192.82   Mean    :272.059
## 3rd Qu.:269.69   3rd Qu.:411.02   3rd Qu.:204.82   3rd Qu.:302.440
```

##	Max.	:2324.60	Max.	:3041.96	Max.	:2242.75	Max.	:509.466
##	wser		wmfg		wfed		wsta	
##	Min.	: 1.844	Min.	:101.8	Min.	:255.4	Min.	:173.0
##	1st Qu.:	191.319	1st Qu.:	234.0	1st Qu.:	361.5	1st Qu.:	258.2
##	Median :	216.475	Median :	271.6	Median :	404.0	Median :	289.4
##	Mean :	224.671	Mean :	285.2	Mean :	403.9	Mean :	296.9
##	3rd Qu.:	247.155	3rd Qu.:	320.0	3rd Qu.:	444.6	3rd Qu.:	331.5
##	Max.	:2177.068	Max.	:646.9	Max.	:598.0	Max.	:548.0
##	wloc		mix		pctymle		lcrmrte	
##	Min.	:163.6	Min.	:0.002457	Min.	:0.06216	Min.	: -6.314
##	1st Qu.:	226.8	1st Qu.:	0.075324	1st Qu.:	0.07859	1st Qu.:	-3.998
##	Median :	253.1	Median :	0.102089	Median :	0.08316	Median :	-3.560
##	Mean :	258.0	Mean :	0.139396	Mean :	0.08897	Mean :	-3.609
##	3rd Qu.:	289.3	3rd Qu.:	0.149009	3rd Qu.:	0.08919	3rd Qu.:	-3.260
##	Max.	:388.1	Max.	:4.000000	Max.	:0.27436	Max.	: -1.809
##	lprbarr		lprbconv		lprbpris		lavgsen	
##	Min.	: -2.833	Min.	: -2.6827	Min.	: -1.9042	Min.	:1.440
##	1st Qu.:	-1.524	1st Qu.:	-1.0564	1st Qu.:	-0.9824	1st Qu.:	:1.969
##	Median :	-1.279	Median :	-0.7458	Median :	-0.8473	Median :	:2.139
##	Mean :	-1.274	Mean :	-0.6929	Mean :	-0.8786	Mean :	:2.153
##	3rd Qu.:	-1.043	3rd Qu.:	-0.4532	3rd Qu.:	-0.7273	3rd Qu.:	:2.322
##	Max.	: 1.012	Max.	: 3.6109	Max.	: -0.3878	Max.	:3.252
##	lpolpc		ldensity		lwcon		lwtuc	
##	Min.	: -7.688	Min.	: -1.62091	Min.	:4.184	Min.	:3.362
##	1st Qu.:	-6.733	1st Qu.:	-0.62934	1st Qu.:	:5.307	1st Qu.:	:5.761
##	Median :	-6.536	Median :	-0.04857	Median :	:5.466	Median :	:5.881
##	Mean :	-6.491	Mean :	-0.01593	Mean :	:5.463	Mean :	:5.916
##	3rd Qu.:	-6.318	3rd Qu.:	: 0.41066	3rd Qu.:	:5.597	3rd Qu.:	:6.019
##	Max.	: -3.336	Max.	: 2.17789	Max.	:7.751	Max.	:8.020
##	lwtrd		lwfir		lwser		lwmfg	
##	Min.	:2.826	Min.	:1.257	Min.	:0.6118	Min.	:4.623
##	1st Qu.:	:5.124	1st Qu.:	:5.463	1st Qu.:	:5.2539	1st Qu.:	:5.455
##	Median :	:5.223	Median :	:5.578	Median :	:5.3775	Median :	:5.604
##	Mean :	:5.232	Mean :	:5.579	Mean :	:5.3646	Mean :	:5.615
##	3rd Qu.:	:5.322	3rd Qu.:	:5.712	3rd Qu.:	:5.5100	3rd Qu.:	:5.768
##	Max.	:7.715	Max.	:6.233	Max.	:7.6857	Max.	:6.472
##	lwfed		lwsta		lwloc		lpctymle	
##	Min.	:5.543	Min.	:5.153	Min.	:5.097	Min.	: -2.778
##	1st Qu.:	:5.890	1st Qu.:	:5.554	1st Qu.:	:5.424	1st Qu.:	: -2.543
##	Median :	:6.001	Median :	:5.668	Median :	:5.534	Median :	: -2.487
##	Mean :	:5.989	Mean :	:5.678	Mean :	:5.540	Mean :	: -2.443
##	3rd Qu.:	:6.097	3rd Qu.:	:5.804	3rd Qu.:	:5.667	3rd Qu.:	: -2.417
##	Max.	:6.394	Max.	:6.306	Max.	:5.961	Max.	: -1.293
##	lpctmin		ltaxpc		lmix			
##	Min.	:0.2497	Min.	:2.660	Min.	: -6.009		
##	1st Qu.:	:2.3030	1st Qu.:	:3.154	1st Qu.:	: -2.586		
##	Median :	:3.2127	Median :	:3.325	Median :	: -2.282		
##	Mean :	:2.9134	Mean :	:3.356	Mean :	: -2.234		
##	3rd Qu.:	:3.6434	3rd Qu.:	:3.505	3rd Qu.:	: -1.904		
##	Max.	:4.1643	Max.	:4.786	Max.	: 1.386		

Q2b (5 points)

Calculate the mean, variance and median of probability of conviction (`prbconv`) by omitting the missing values, if any.

```
# Mean of prbconv (remove NA values)
mean_prbconv <- mean(Crime$prbconv, na.rm = TRUE)

# Variance of prbconv (remove NA values)
var_prbconv <- var(Crime$prbconv, na.rm = TRUE)

# Median of prbconv (remove NA values)
median_prbconv <- median(Crime$prbconv, na.rm = TRUE)

# Print the results
cat("Mean:", mean_prbconv, "\n")
```

```
## Mean: 0.6886177
```

```
cat("Variance:" , var_prbconv , "\n" )
```

```
## Variance: 2.857265
```

```
cat("Median:", median_prbconv, "\n")
```

```
## Median: 0.4743745
```

Q2c (10 points)

Use `lpolpc` (log-police per capita) and `smsa` variables to build a linear regression model to predict probability of conviction (`prbconv`). And, compare with another linear regression model that uses `polpc` (police per capita) and `smsa`.

```
# Comparing lpolpc and smsa using a linear regression model
modelone <- lm(prbconv ~ lpolpc + smsa, data = Crime)
# Summarize the model
summary(modelone)
```

```
##
## Call:
## lm(formula = prbconv ~ lpolpc + smsa, data = Crime)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.183 -0.418 -0.134  0.176 33.593
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.5788     0.7877  10.891 < 2e-16 ***
## lpolpc         1.2049     0.1204  10.005 < 2e-16 ***
```



```
## smsayes      -0.7827      0.2227  -3.514 0.000472 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.568 on 627 degrees of freedom
## Multiple R-squared:  0.1418, Adjusted R-squared:  0.1391
## F-statistic:  51.8 on 2 and 627 DF,  p-value: < 2.2e-16
```

```
# Comparing polpc and smsa using a linear regression model
modeltwo <- lm(prbconv ~ polpc + smsa, data = Crime)
# Summarize the model
summary(modeltwo)
```

```
##
## Call:
## lm(formula = prbconv ~ polpc + smsa, data = Crime)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.608 -0.242 -0.094  0.090 32.998
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.19444    0.07555   2.574   0.0103 *
## polpc         278.52397   21.97066  12.677   <2e-16 ***
## smsayes       -0.44656    0.21098  -2.117   0.0347 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.507 on 627 degrees of freedom
## Multiple R-squared:  0.2078, Adjusted R-squared:  0.2053
## F-statistic: 82.25 on 2 and 627 DF,  p-value: < 2.2e-16
```

How can you draw a conclusion from the results of part Q2c? (Note: Full marks requires comment on the predictors)

```
# comment: Model 2, which uses polpc and smsa, is a much better model for predicting the probability of
```

Q2d (10 points)

Based on the output of your model, predict the output for different factors of `smsa` when `polpc` is set to 0.0025. Use both the `predict` function and manual intercept and slope equations and compare the results.

```
# Create a data frame for prediction using the 'predict' function
new_data <- data.frame(polpc = 0.0025, smsa = c("no", "yes"))

# Predict using the model
predicted_values <- predict(modeltwo, newdata = new_data)

# Print the predicted values
predicted_values
```

```
##           1           2
## 0.8907482 0.4441882
```

```
# Manual Option
# Define the values
intercept <- 0.19444
polpc_coefficient <- 278.52397
smsayes_coefficient <- -0.44656
polpc_value <- 0.0025

# For rural areas (smsa = "no")
prbconv_rural <- intercept + (polpc_coefficient * polpc_value) + 0 # Rural (smsa = "no")

# For urban areas (smsa = "yes")
prbconv_urban <- intercept + (polpc_coefficient * polpc_value) + smsayes_coefficient # Urban (smsa = "yes")

cat("Rural predicted prbconv:" ,prbconv_rural , "\n" )
```

```
## Rural predicted prbconv: 0.8907499
```

```
cat("Urban predicted prbconv:" , prbconv_urban , "\n")
```

```
## Urban predicted prbconv: 0.4441899
```

Q2e (10 points)

Find Pearson correlation between probability of prison sentence `prbpris` and tax per capita `taxpc`; and also Pearson correlation between probability of conviction `prbconv` and probability of arrest `prbarr`.

```
# Pearson correlation between prbpris and taxpc
cor_prbpris_taxpc <- cor(Crime$prbpris, Crime$taxpc, use = "complete.obs")

# Pearson correlation between prbconv and prbarr
cor_prbconv_prbarr <- cor(Crime$prbconv, Crime$prbarr, use = "complete.obs")

cat("Pearson correlation between prbpris and taxpc:", cor_prbpris_taxpc, "\n")
```

```
## Pearson correlation between prbpris and taxpc: -0.1120631
```

```
cat("Pearson correlation between prbconv and prbarr:", cor_prbconv_prbarr, "\n")
```

```
## Pearson correlation between prbconv and prbarr: 0.0355689
```

What conclusions can you draw after using `cor.test`? Write your reasons as comments.

```
# comment:The correlation coefficient between 'prbpris' and 'taxpc' is 0.45, indicating a somewhat posi
```

Q2f (10 points)

Display the correlation matrix of the variables: `prbconv`, `prbpris`, `density`, `polpc`.

```

# Select the relevant variables
selected_vars <- Crime[, c("prbconv", "prbpris", "density", "polpc")]
# Compute the correlation matrix
cor_matrix <- cor(selected_vars, use = "complete.obs")
# Display the correlation matrix
print(cor_matrix)

```

```

##           prbconv    prbpris    density    polpc
## prbconv  1.00000000 -0.03734017 -0.11555529  0.44963500
## prbpris -0.03734017  1.00000000  0.16466324 -0.05745238
## density -0.11555529  0.16466324  1.00000000 -0.03969574
## polpc    0.44963500 -0.05745238 -0.03969574  1.00000000

```

Write what conclusion you can draw, as comments.

```

# comment: Based on the correlation matrix, there is a somewhat positive association between the prbconv

```

Question 3 (25 points)

Many times in data analysis, we need a method that relies on repeated random sampling to obtain numerical results. The underlying concept is to use randomness to solve problems. In fact, this is a mathematical technique, which is used to estimate the possible outcomes of an uncertain event and is called the Monte Carlo Method.

Consider that we roll a die 10 times and want to know the probability of getting more than 3 times of even numbers. This is a problem for the Binomial distribution, but suppose we don't know anything about Binomial distribution. We can easily solve this problem with a simulation or random number generation, called Monte Carlo Simulation.

Q3a (5 points)

The Monte Carlo Method uses random numbers to simulate some process. Here, the process is rolling a die 20 times. Assume the die is fair. What is the probability of success or getting an even number in rolling the die once?

```

# There are 3 even numbers (2, 4, 6) out of 6 possible outcomes.
prob_success <- 3 / 6
# Print the probability
cat("The probability of getting an even number in one roll is:", prob_success, "\n")

```

```

## The probability of getting an even number in one roll is: 0.5

```

Q3b (10 points)

Define a function named `one_trial`, that simulates a single round of rolling a die 20 times and returns true if the number of even numbers is > 5 .

```

one_trial <- function() {
  # Simulate rolling a die 20 times
  rolls <- sample(1:6, size = 20, replace = TRUE)

  # Count the number of even numbers
  num_even <- sum(rolls %% 2 == 0)

  # Return TRUE if the count of even numbers is greater than 5, otherwise FALSE
  return(num_even > 5)
}

# Example: Run one trial
result <- one_trial()
cat("Result of one_trial:", result, "\n")

```

```
## Result of one_trial: TRUE
```

Q3c (5 points)

Repeat the function `one_trial` for $N = 100,000$ times and sum up the outcomes and store the result in a variable named `desired_output`. Compute the probability of getting more than 5 times of even numbers by using relative frequency.

```

# Set the number of trials
N <- 100000

# Repeat one_trial N times and store the results in a logical vector
trial_results <- replicate(N, one_trial())

# Sum the outcomes (TRUE is treated as 1, FALSE as 0)
desired_output <- sum(trial_results)

# Compute the probability using relative frequency
estimated_probability <- desired_output / N

# Print the results
cat("The estimated probability of getting more than 5 even numbers in 20 rolls is:",
    estimated_probability, "\n")

```

```
## The estimated probability of getting more than 5 even numbers in 20 rolls is: 0.97916
```

Q3d (5 points)

Use the Binomial formula you learned before to calculate such probability and Compare it with the probability value obtained in part (c).

```

# Calculate the probability of getting more than 5 even numbers in 20 rolls using the Binomial formula
binom_prob <- 1 - pbinom(5, size = 20, prob = 0.5)
cat("The probability using the Binomial formula is:", binom_prob, "\n")

```

```
## The probability using the Binomial formula is: 0.9793053
```

Congratulations! You have completed the first run of the Monte Carlo simulation.

If there is further interest, put all the above logic in a function, call it at least 50 times, store the results in a vector called Prob, and, to be more accurate, take the mean of the Prob vector. (optional)

End of Assignment